Chapter

4

ENGINEERING MATERIALS

SIF2TD

Learning Objectives

1. Students to know about the various Engineering Materials and their properties and their types.
2. To know about the metals and non-metals and Ferrous and non-ferrous metals, steels and cast Iron.
3. To know, the uses of all Engineering materials.

Table of Content

4.1 Introduction
4.2. Engineering Material
4.3. Properties of Materials – Physical Properties – Chemical Properties – Electrical Properties – Mechanical Properties
4.4. Metals
Ferrous Metals – (Pure Iron – Steel – Cast Iron)
Non Ferrous Metals – (Aluminium – Copper – Brass – Lead – Zinc)

4.1 Introduction

- In General, material may be anything else. It is consisting of whether pure or impure, single or composite, living or non-living matter. It may be a solid, liquid, and gas. Materials can be classified based on different properties, such as physical and chemical properties, etc.,
- In Industry, Material are inputs for production or manufacturing processes. They may be raw material to produce things in engineering fields. Synthetic materials are used in other fields like Medical, Textile and Home Appliances. Medicines, textile, petrol, fuels, soaps, glass, polymers, cements, etc are synthetic materials.
- The materials are either natural or artificial. Natural materials are air, water, oil, minerals, cereals, metals, wood, rubber, etc.,
- The artificial materials are made from natural things. In the physical sense, materials are studied in material science.

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